5.2 The Mean Value Theorem

AP Calculus 5.2 · Student sheet

Name _____ Period ____ Date _____

I can…

- use Rolle's Theorem to find c with f′(c) = 0.
- find the number c guaranteed by the Mean Value Theorem.
- find where a function is increasing and decreasing.

Vocabulary · fill in the missing word

Rolle's Theorem — if f is continuous on [a, b], _____ on (a, b), and f(a) = f(b), then f′(c) = 0 for some c in (a, b)

Mean Value Theorem — if f is continuous on [a, b] and _____ on (a, b), then $f'(c) = \frac{f(b) - f(a)}{b - a}$ for some c in (a, b)

Increasing/Decreasing Test — f′ > 0 on an _____ means f is increasing; f′ < 0 means f is decreasing

Part 1 · Speeding ticket

120 miles in 2 hours.

Question	Answer
Average speed	
Was the speed exactly 60 mph at some instant?	
Why?	

Part 2 · Notes and examples (fill in each note, then work the example)

Rolle's Theorem: if f is continuous on [a, b], differentiable on (a, b), and f(a) = f(b), then f′(c) = _____ for some c in (a, b).

Example 1

Verify Rolle's Theorem for $f(x) = x^2 - 4x$ on [0, 4] and find c.

Mean Value Theorem: under the same first two conditions, f′(c) = _____ for some c in (a, b).

Example 2

Find c guaranteed by the Mean Value Theorem for $f(x) = x^2$ on [1, 3].

Example 3

Find c for $f(x) = x^3$ on [0, 3].

Increasing/Decreasing Test: if f′(x) > 0 on an interval, then f is _____ on that interval.

Example 4

Where is $f(x) = x^3 - 3x^2 - 9x$ increasing? Decreasing?

Example 5

$f(1) = 3$ and $f(5) = 11$ (f continuous and differentiable). What value must f' take?

Part 3 · Partner practice

Partner 1

Where is $f(x) = x^3 - 12x$ increasing?

Partner 2

(AP® Practice) f is continuous on $[1, 4]$ and differentiable on $(1, 4)$, with $f(1) = -2$ and $f(4) = 7$. Which must be true?

(A) $f'(c) = 3$ for some c in $(1, 4)$ (B) $f'(c) = 0$ for some c in $(1, 4)$

(C) $f(c) = 8$ for some c in $(1, 4)$ (D) $f'(x) > 0$ for all x in $(1, 4)$